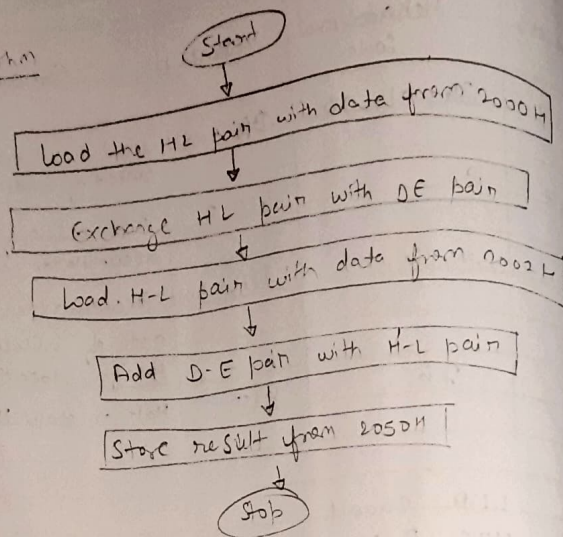


Algorithm



Aim:- Addition of two 16 bit numbers using 16-bit operation code on microprocessor 8085.

Apparatus:- Power supply 8085 microprocessor kit

Algorithm:-

1. Store the value at a successive memory location
2. Start the program from a particular address.
3. Load both the lower & higher bits of the first number at once in the register pair.
4. Copy the first number to another register pair.
5. Load both the lower & higher bits of the second number at once in the register pair.
6. Add both the register pairs & store the result in a memory location.
7. Stop the program.

Program Hexadecimal

Address	Code	Memories	Comment
2010	2A, 00, 20	LHLD, 2000H	load the value of 2000H in L register & that of 2001 in H register
2013	EB	XCH	Copy the content of H to D & L to E reg.
2014	2A, 02, 20	LHLD, 2002H	load the values of 2002 in L register & that of 2003 in H register.

Output

Data used

Address	Data
12	
2000	34
2001	45
2002	67
2003	

Result

Address	Result
2050	57
2051	98

Date _____

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Address	Hexadecimal code	Memories	Comments
2017	19	DAD, D	Adds the values of H with D & L with C & stores the result in H & L
2018	22, 50,	SHLD, 2050H	Store result at 2050 hex
	20	HE	
2019	76	HLT	Stop program

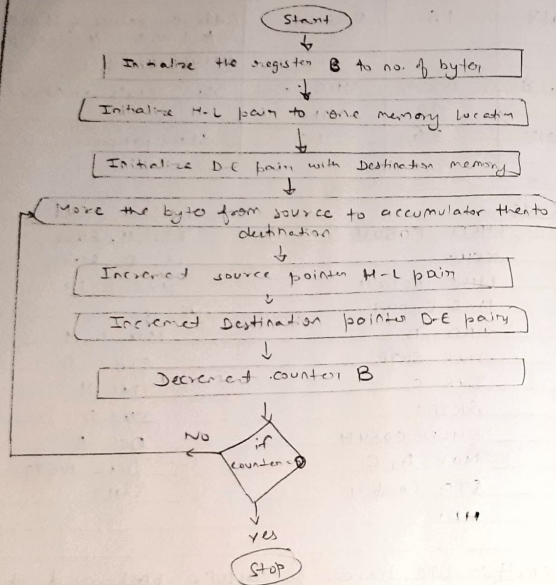
Code

LHLD C050H	LXI H, ED50
XCHG	LXI D, F07D
LHLD C052H	MVI B, 10
MVI C, 00H	NEXT:
DAD D	MOV A, M
JNC SKIP	STAX D
INR C	INX H
SKIP:	INX D
SHLD C054H	OCR B
MOV A, C	JNZ NEXT
STA C056H	HLT
HLT	

Result:- We have successfully performed the addition of 2 16 (bit) data stored at some memory location on the 8085 microprocessor kit & store the result in the successive memory location.

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Algorithm Flow chart



Expt. No. 4.1

Date _____

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Prior: Transfer a block of data from memory location $xx00$ to another location $xx00$ in the direct order.

Apparatus: - 15 volt SPS, 8085 microprocessor kit.

Algorithm:

1. Store the value (data) at a successive memory location.
2. Start the program from a particular address.
3. Load register pair H-L with the address $2000H$.
4. Load register pair D-E with the address $3000H$.
5. Move the content of memory location into accumulator.
6. Move store the content of accumulator into memory location pointed by D-E.
7. Increment value of register pair H-L & D-E by 1.
8. Decrement value of register B by 1.
9. If ~~here~~ zero flag is not equals to 1, go to step 3.
10. Stop.

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Address	Hexadecimal Code	Mnemonic	Comments
2200	DE, 0A	MVI B, 0AH	Move data 0AH to register B immediately
2202	21, 00, 20	LXI H, 2000H	load H-L register pair immediately by 2000H
2205	21, 00, 30	LXI H, 3000H	load D-E register pair immediately by 3000H
2208	7E	Mov A, M	transfer data from memory to accumulator.
2209	12	STAX, D	store accumulator into register pair D-E immediately
220A	23	INX H	increase H-L pair by 1
220B	13	INX D	increase D-E pair by 1
220C	0P	DCR B	decrease register B by 1
220D	C2, 08, 22	JNZ 2208	Jump if not zero to given memory location
2210	76	HLT	Halt the program

Code
 LXI H, 2000
 LXI D, 3000
 MVI B, 0A
 MOV A, M
 STAX D
 INX H
 INX D
 DCR B
 JNZ NEXT
 HLT

Input

Address	Data
2000	0A
2001	01
2002	60
2003	A2
2004	91
2005	08
2006	63
2007	B6
2008	D1
2009	F6

Outputs

Address	Data
3000	0A
3001	01
3002	60
3003	A2
3004	91
3005	08
3006	63
3007	B6
3008	D1
3009	F6

Date _____

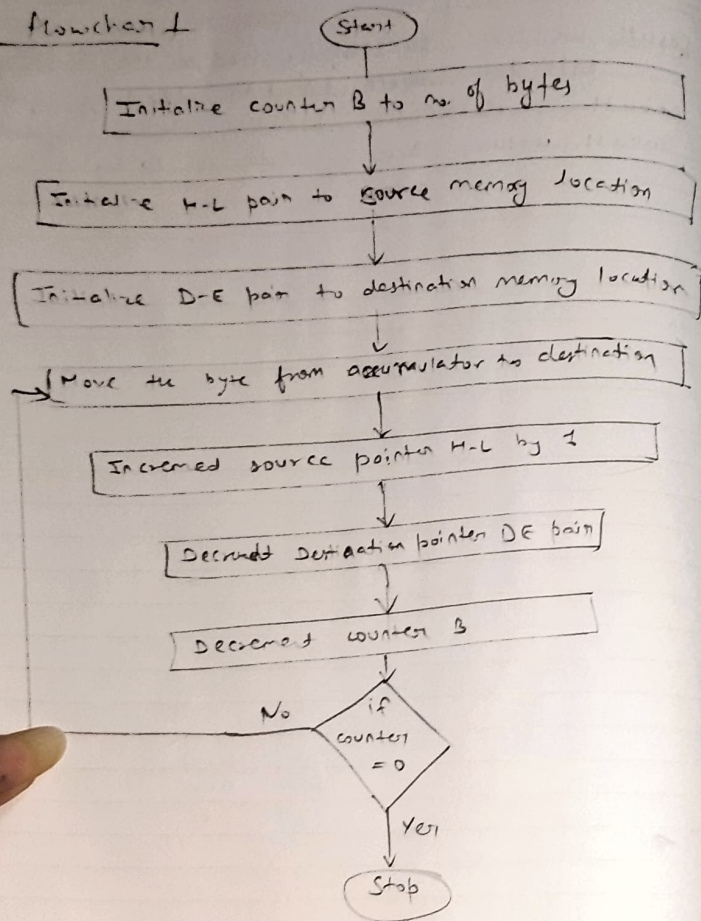
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Result we have successfully moved the block of bits from source location starting from 2000H to destination location starting from 3000H returns size of block @ 16 bytes.

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Flowchart 1



Expt. No. 4:2

Date

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Aim:- Program to move blocks of bits in reverse order from source location to destination location.

Apparatus:- Power supply, 8085 microprocessor kit

Algorithm:-

1. Store the value (data) at a successive memory location.
2. Start the program from a particular address.
3. Load register pair H-L with the address 2000H.
4. Load register pair D-E with the address 3000H.
5. Move the content at memory location into accumulator.
6. Store the content of accumulator into location pointed by DE.
7. Increment value of register pair H-L by 1.
8. Decrement value of register pair DE & register B by 1.
9. If zero flag not equal to 1, go to step 3.
10. Stop.

Program

Address	Hexadecimal Code	Memorics	Comments
2200	0F, 0A	MVI, B, 0AH	Move data 0AH to register B immediately
2202	21, 00, 20	LXI H, 2000H	Load H-L pair immediately by 2000H (16-bit)
2005	11, 00, 30	LXI D, 3000H	Load D-E pair immediately by 3000H (16-bit)

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Input	
Address	Data
2000	00
2001	AD
2002	61
2003	A1
2004	C0
2005	DA
2006	BC
2007	A5
2008	C1
2009	F5

Output	
Address	Data
2000	00
2001	AD
2002	61
2003	A1
2004	C0
2005	DA
2006	BC
2007	A5
2008	C1
2009	F5

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Address	Hexadecimal Code	Memories	Comments	
2208	7E	MOV A, M	Transfer data from memory to accumulator	
2209	12	STAX, D	store accumulator into register pair D immediately	
220A	23	INX H	increase H-L register by 1	
220B	15	DCX D	decrease D-E pair by 1	
220C	00	DCR B	decrease counter by 1	
220D	C2, 08, 22	JNZ, 2208H	Jump if not zero to given memory location	
2210	76	HLT	Halt the program	
<u>Code</u>				
MVI B, 0AH				
LXI H, 2000H				
LDI D, 2000H				
MOV A, M				
STAX, D				
INX H				
DCX D				
DCR B				
JNZ 2208H				
HLT				
<u>Result</u> We have successfully moved the blocks of bit in reverse order from 2000H location containing data starting from 2000H to destination location starting from 2000H where size of block is 16 bits.				
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